

1. Project Overview

This project focuses on an Adaptive Precision Reconfigurable TinyML CNN Accelerator for edge AI applications. The design supports multiple precision modes to balance classification accuracy, memory requirement, computation, and FPGA resource usage.

2. Objectives

- Develop a lightweight CNN for gesture recognition.
- Evaluate 8-bit, 4-bit, and 1-bit quantized models.
- Design a reconfigurable accelerator with selectable precision.
- Implement and verify the design on a Xilinx Artix-7 XC7A35T FPGA.
- Analyze resource utilization, timing, and power.

3. Methodology

The CNN model was developed and quantized using TensorFlow/TensorFlow Lite. The 8-bit, 4-bit, and 1-bit models were evaluated for accuracy and model size. Based on the results, the hardware architecture was designed with 1-bit, 4-bit, and 8-bit precision modes.

The accelerator contains Input Memory, Window Generator, Precision Controller, Adaptive MAC Unit, ReLU, Max Pooling, and Output Register. The design was implemented and verified using Xilinx Vivado.

4. Quantization Results

Precision	Accuracy	Model Size	Observation
8-bit	100.00%	812 KB	Highest accuracy
4-bit	99.94%	406 KB	Best balance
1-bit	28.25%	101 KB	Lowest memory, lower accuracy

The 4-bit model was selected for hardware implementation because it achieved 99.94% accuracy with a reduced model size.

5. Hardware Implementation

The proposed architecture was implemented on the Xilinx Artix-7 XC7A35T FPGA. The adaptive MAC unit performs the main multiplication and accumulation operations, while the precision controller selects the operating precision.

Parameter	Result	Utilization / Status
Slice LUTs	175	0.84%
Slice Registers	68	0.16%
Slices	63	0.77%
DSPs	0	0%
BRAM	0	0%
Operating Frequency	100 MHz	Timing met
Estimated Maximum Frequency	121.2 MHz	—

Total On-Chip Power	79 mW	—
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6. Timing and Power

Initial timing issues caused by long combinational paths were reduced by adding pipeline registers between processing stages. After pipelining, the design satisfied the 100 MHz clock requirement. The reported WNS is +1.751 ns and WHS is +0.070 ns.

The total estimated on-chip power is 79 mW, including 60 mW static power and 18 mW dynamic power.

7. Work Completed

- CNN development and dataset evaluation.
- 8-bit, 4-bit, and 1-bit quantization experiments.
- Selection of 4-bit precision for hardware implementation.
- Adaptive precision accelerator architecture design.
- Verilog development and behavioral verification.
- FPGA synthesis, resource, timing, and power analysis.
- Comparison with existing CNN accelerator designs and conference-paper preparation.

8. Conclusion

The proposed accelerator provides a compact FPGA-based solution for TinyML edge applications. The 4-bit model achieved 99.94% classification accuracy, while the hardware used 175 LUTs and 68 registers, operated at 100 MHz, and showed an estimated power consumption of 79 mW.

9. Future Scope

- Support additional CNN layers and networks.
- Explore more precision modes and dynamic switching.
- Evaluate other low-power FPGA devices.
- Further optimize power and memory for edge deployment.

References

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